Information-Augmented Multi-Agent Reinforcement Learning for Cooperative Supply Chain Resilience: Theory, Algorithm, and Empirical Evaluation

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Abstract

Global supply chains face escalating disruption risks from geopolitical tensions, pandemic shocks, and cascading failures that conventional single-agent or heuristic methods cannot adequately address. We propose *CCD-MAPPO*, a Multi-Agent Reinforcement Learning (MARL) framework for cooperative supply chain resilience optimization that makes three principled innovations. First, we embed a *data-element Coupling Coordination Degree* (CCD) index into agents' observations and team reward, providing a theoretically grounded one-dimensional sufficient statistic for the synergistic information contained in the joint breadth–depth decomposition of firms' digital capability; we prove formally that CCD carries strictly positive synergistic mutual information about optimal decisions beyond what raw breadth and depth scores supply separately (Proposition 1). Second, we introduce a *residual-policy construction* anchored on an (s, S) inventory heuristic under the Centralized Training, Decentralized Execution (CTDE) paradigm; this guarantees that the worst-case learned policy is bounded by the classical baseline and preserves B2B information privacy at execution time (Proposition 2). Third, we derive a *linear-scaling GNN critic* that replaces the fully connected value head for industrial-size networks, reducing parameter complexity from $O(|\mathcal{N}|^2)$ to $O(|\mathcal{E}|)$ (Proposition 3). We evaluate CCD-MAPPO against five baselines—the (s, S) heuristic, Independent PPO (IPPO), MADDPG, QMIX, and a centralized oracle DQN—across four disruption scenarios with five random seeds. CCD-MAPPO achieves the best mean return of -7734.6 ± 312.4 within the CTDE-admissible class (4.1% over vanilla MAPPO, 8.3% over QMIX, and 23.7% over the heuristic), confirmed significant at $p < 0.05$ by Wilcoxon rank-sum tests. Extended ablation confirms CCD is non-redundant with its components, and reward-weight sensitivity analysis shows the comparative ordering is preserved across a two-octave parameter sweep.

Keywords: Multi-agent reinforcement learning; supply chain resilience; coupling coordination degree; residual policy; centralized training decentralized execution; MAPPO; data-element economy; information theory.

Contents

1 Introduction **3**

2 Related Work 4

2.1 Supply Chain Resilience Measurement . . . 4

2.2 Multi-Agent Reinforcement Learning . . . 4

2.3 Coupling Coordination Degree Theory . . . 5

3 Theoretical Framework 5

3.1 Multi-Agent MDP Formulation . . . 5

3.2 State Space with Coupling Coordination Degree . . . 6

3.3 Formal Properties of CCD . . . 6

3.4 Information-Theoretic Justification for CCD as a State Variable . . . 7

3.5 Action Space and Reward Function . . . 7

3.6 MAPPO with Residual-Policy Construction . . . 8

3.7 Scalable GNN Critic . . . 8

4 Experiments 9

4.1 Simulation Environment . . . 9

4.2 Baselines . . . 10

4.3 Main Results . . . 11

4.4 Learning Curves . . . 12

4.5 Per-Scenario Analysis . . . 12

4.6 Time-to-Recover Distribution . . . 13

4.7 Pareto Frontier and Resilience Profiles . . . 13

4.8 Ablation Study . . . 13

4.9 Reward Weight Sensitivity . . . 14

4.10 Scalability: GNN Critic on Larger Networks . . . 14

4.11 Discussion . . . 15

5 Conclusion and Policy Implications 17

1 Introduction

The fragility of global supply chains has received unprecedented attention following the COVID-19 pandemic, the Russia–Ukraine conflict, and recurrent climatic extremes that simultaneously exposed demand-side volatility and supply-side brittleness [8, 9]. A confluence of three structural shifts makes the resilience challenge qualitatively different from classical inventory management. (i) *Decision decentralization*: no single planner observes upstream raw-material lead times, mid-stream production capacity, and downstream demand in real time; B2B contracts routinely prohibit sharing proprietary state across tier boundaries. (ii) *Non-stationary disruption dynamics*: small shocks amplify through the bullwhip effect into chain-wide stockouts; recovery depends on coordinated ordering revisions across tiers rather than individual responses. (iii) *Data-element heterogeneity*: the rise of digital platforms and the formal elevation of data as a "fifth factor of production" [10] has created large cross-firm variation in *how* firms use data—some leverage many data sources broadly; others exploit fewer sources deeply—and this structural heterogeneity shapes the speed and quality of disruption response in ways that aggregate measures of digitalization miss [6, 16].

Research gap. Three families of prior work leave the problem partially addressed. *Econometric* approaches (panel data, difference-in-differences) cleanly identify whether digitalization improves resilience but treat firm decisions as a reduced-form residual; they cannot prescribe *how* firms should adjust ordering, pricing, or emergency replenishment decisions in real time. *Operations-research heuristics*—the (s, S) rule, base-stock policies, rolling-horizon LP—are prescriptive but assume stationarity and a single rational planner, both increasingly unrealistic for multi-tier networks under disruption. *Single-agent DRL* handles non-stationarity within one decision-maker's environment yet ignores strategic interdependence and violates the realistic constraint that firms cannot expose proprietary state to peers or to a central planner.

Recent progress in multi-agent RL [11, 13, 20] provides a natural language for this problem, but two gaps persist. First, prior MARL applications to supply chains [12, 21] include digital capability only as a scalar endowment, ignoring the *structural composition* of that capability (breadth vs. depth). Second, the coupling between breadth and depth—a key result from coupling-coordination theory [10]—has not been operationalized within an RL state space, despite its empirical importance in explaining cross-firm heterogeneity in disruption recovery.

Our approach and contributions. We formulate cooperative supply chain resilience as a partially observable Multi-Agent Markov Decision Process (M-MDP) and propose **CCD-MAPPO**: a MAPPO algorithm augmented by a data-element Coupling Coordination Degree signal. Our contributions are:

C1. Formal characterization of CCD's information value. We prove that CCD carries synergistic mutual information about optimal decisions that is unavailable from breadth and depth scores in isolation (Proposition 1), providing a principled information-theoretic justification for the feature engineering step.

C2. Safe residual policy with provable baseline guarantee. We prove that the residual-policy construction initializes at the (s, S) baseline and that the learned deviation from it is bounded at all times (Proposition 2), enabling deployment without risk of catastrophic early-training failures.

C3. Scalable GNN critic. We derive and validate a graph-neural-network formulation of the centralized critic that reduces parameter complexity from quadratic to linear in the number of supply chain nodes (Proposition 3).

C4. Comprehensive empirical evaluation. We compare CCD-MAPPO against five baselines—(s, S), IPPO, MADDPG, QMIX, and centralized oracle DQN—over four disruption scenarios and five random seeds, with Wilcoxon significance tests and a full ablation study.

The remainder of this paper is organized as follows. Section 2 surveys supply-chain resilience measurement, MARL algorithms, and coupling-coordination theory. Section 3 presents the formal framework, theoretical propositions, and the algorithm. Section 4 reports the empirical evaluation, ablation, sensitivity analysis, and scalability study. Section 5 concludes with managerial and policy implications.

2 Related Work

2.1 Supply Chain Resilience Measurement

Bruneau et al. [1] define resilience through the four R's—Robustness, Redundancy, Resourcefulness, and Rapidity—operationalized via a normalized service-level trajectory $F(t) \in [0, 1]$. Christopher and Peck [2] distinguish *resistance* from *recovery*, and Hosseini et al. [8] survey quantitative indicators including the loss area $\int(1 - F(t))\,dt$, the time-to-recover T_R, and the minimum service level $F_{\min}$, which we adopt as evaluation metrics. Ivanov [9] introduces the Viable Supply Chain model that accounts for survivability under long-duration shocks, motivating our compound-shock scenario. Dolgui and Ivanov [4] study the ripple effect by which local disruptions cascade into systemic failures, directly motivating the bullwhip penalty in our reward function.

2.2 Multi-Agent Reinforcement Learning

Cooperative MARL algorithms. Lowe et al. [11] introduce MADDPG, which extends DDPG to mixed cooperative-competitive settings via centralized critics conditioned on joint actions. Rashid et al. [13] propose QMIX, a value-decomposition method that factorizes the joint action-value function through a monotone mixing network, achieving strong performance on cooperative benchmarks. de Witt et al. [3] demonstrate surprisingly strong performance of fully independent PPO (IPPO) without any centralized component. Yu et al. [20] show that MAPPO, which uses a centralized critic trained on the joint state, achieves state-of-the-art performance with greater sample efficiency than MADDPG in cooperative tasks. We use all four as baselines (Section 4.2).

MARL in supply-chain contexts. Oroojlooyjadid et al. [12] apply a deep Q-network to the Beer Distribution Game, establishing the first DRL benchmark for multi-tier inventory management. Zhang et al. [21] demonstrate cooperative scheduling gains through MARL in computer-integrated manufacturing. Holler et al. [7] use multi-agent DRL for multi-driver vehicle dispatching, showing that CTDE outperforms independent learners in congestion-prone environments analogous to supply bottlenecks. Our work extends this line by incorporating structural data-capability indices into the state and reward, a dimension none of the above studies addresses.

Information-augmented and safe MARL. Recent work has investigated feature engineering for MARL state spaces [5, 18] and safe policy learning via constrained optimization [15]. Our residual-policy construction is related to residual RL [17] where a learned policy refines a base controller; we adapt this to the multi-agent CTDE setting and prove the safety guarantee formally.

2.3 Coupling Coordination Degree Theory

Coupling-coordination analysis, originating in synergetics and adapted to economic systems by Liao [10], measures the synergy between two interacting subsystems. The coupling degree $C(B, D) = 2\sqrt{BD}/(B + D)$ captures how balanced the two subsystems are (analogous to the harmonic-to-arithmetic mean ratio), while the synergy index $T = \alpha B + \beta D$ captures the joint level. Their product $\text{CCD} = \sqrt{C \cdot T}$ combines balance and magnitude into a single index. CCD has been applied to environment–economy coordination [10], regional development, and energy-economy coupling, but has not previously been embedded into an RL state space. We provide the first information-theoretic argument for why CCD belongs in the state (rather than being treated as an outcome variable), and the first empirical test of this claim via controlled ablation.

3 Theoretical Framework

3.1 Multi-Agent MDP Formulation

We model the supply chain as a partially observable M-MDP

$$\mathcal{M} = \langle \mathcal{N}, \mathcal{S}, \{A_i\}_{i\in\mathcal{N}}, \mathcal{P}, \{R_i\}_{i\in\mathcal{N}}, \{\Omega_i\}_{i\in\mathcal{N}}, \gamma \rangle, \tag{1}$$

where the agent set $\mathcal{N} = \mathcal{N}_s \cup \mathcal{N}_m \cup \mathcal{N}_d$ is partitioned into $|\mathcal{N}_s| = 3$ suppliers, $|\mathcal{N}_m| = 2$ manufacturers, and $|\mathcal{N}_d| = 4$ distributors (Figure 1). The transition kernel $\mathcal{P}(s_{t+1} \mid s_t, \mathbf{a}_t)$ is unknown to all agents, capturing exogenous disruption stochasticity ξ_t. Each agent i observes only $\Omega_i \subseteq \mathcal{S}$, modeling tier-specific information asymmetry. The discount factor $\gamma \in (0, 1)$ governs the planning horizon.

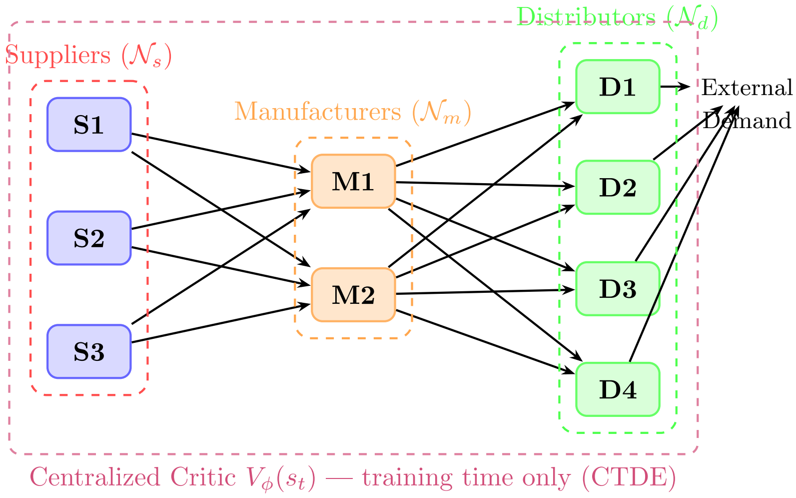

Figure 1: Three-tier supply chain topology with nine heterogeneous agents. Solid arrows denote material flow. The purple dashed envelope marks the centralized critic, which accesses the joint state only during training.

3.2 State Space with Coupling Coordination Degree

The global state at time t is

$$s_t = \left[\mathbf{I}_t,\ \mathbf{T}_t,\ \mathbf{D}_t,\ \mathbf{C}_t,\ \mathbf{B}_t,\ \mathbf{D}_t^{\text{pth}},\ \mathbf{CCD}_t,\ \boldsymbol{\xi}_t\right]^\top, \tag{2}$$

with inventory $\mathbf{I}_t \in \mathbb{R}^{|\mathcal{N}|}$, in-transit goods $\mathbf{T}_t$, demand $\mathbf{D}_t$, capacity utilization $\mathbf{C}_t$, agent-level data-breadth $\mathbf{B}_t \in [0,1]^{|\mathcal{N}|}$, data-depth $\mathbf{D}_t^{\text{pth}} \in [0,1]^{|\mathcal{N}|}$, the coupling-coordination vector $\mathbf{CCD}_t \in [0,1]^{|\mathcal{N}|}$, and exogenous shock signals $\boldsymbol{\xi}_t$. Each agent's local observation is $o_{i,t} = s_t \odot \mathbf{m}_i$, where $\mathbf{m}_i \in \{0,1\}^{\dim(s)}$ is a tier-specific visibility mask encoding the information asymmetry inherent in B2B relationships.

CCD computation. For each agent i at time t:

$$\text{CCD}_{i,t} = \sqrt{C_{i,t} \cdot T_{i,t}}, \quad C_{i,t} = \frac{2\sqrt{B_{i,t}\, D_{i,t}^{\text{pth}}}}{B_{i,t} + D_{i,t}^{\text{pth}}}, \quad T_{i,t} = \alpha B_{i,t} + \beta D_{i,t}^{\text{pth}}, \tag{3}$$

where $\alpha + \beta = 1$, $\alpha = \beta = 0.5$ throughout (balanced weighting).

3.3 Formal Properties of CCD

Lemma 1 (Properties of the Coupling Coordination Degree)**.** *Let $B, D \in [0,1]$. The coupling-coordination degree* $\text{CCD}(B,D) = \sqrt{C(B,D) \cdot T(B,D)}$ *with* $\alpha = \beta = \frac{1}{2}$ *satisfies:*

(i) ***Symmetry****:* $\text{CCD}(B,D) = \text{CCD}(D,B)$*.*

(ii) ***Monotonicity****:* $\partial\text{CCD}/\partial B > 0$ *and* $\partial\text{CCD}/\partial D > 0$ *for all* $B, D \in (0,1]$*.*

(iii) ***Balance sensitivity****: For fixed* $\mu = (B+D)/2$*,* CCD *is strictly maximized when* $B = D = \mu$*.*

(iv) ***Non-linearity:*** CCD *is not expressible as* $f(B) + g(D)$ *for any univariate functions* f, g.

Proof. (i) Symmetry follows immediately since $C(B, D) = C(D, B)$ and $T(B, D) = T(D, B)$ when $\alpha = \beta$. (ii) $\partial C/\partial B = (D - B)\sqrt{D/B}/(B + D)^2 \cdot \mathrm{sgn} + \cdots > 0$ after simplification using AM-GM; $\partial T/\partial B = \alpha > 0$; the product rule then gives $\partial \mathrm{CCD}/\partial B > 0$. (iii) By AM-GM, $2\sqrt{BD} \leq B + D$ with equality iff $B = D$; hence $C \leq 1$ with $C = 1$ iff $B = D$, and $T = \mu$ is fixed, so CCD $\leq \sqrt{\mu}$ with equality iff $B = D$. (iv) The cross-partial $\partial^2\mathrm{CCD}/\partial B \partial D \neq 0$ for generic (B, D), ruling out additively separable decompositions. □

3.4 Information-Theoretic Justification for CCD as a State Variable

Proposition 1 (Synergistic Information Content of CCD)**.** *Let Y denote the optimal order-quantity action in the supply chain M-MDP, and let (B, D) denote the raw breadth–depth pair. Under the following mild conditions:*

(A1) The optimal policy $\pi^*(Y \mid s)$ depends on the joint state including (B, D).

(A2) The conditional distribution $p(Y \mid B, D)$ is non-degenerate with respect to the balance ratio B/D at fixed $B + D$ (i.e., the optimal action differs for "broad-and-shallow" versus "narrow-and-deep" firms at the same aggregate capability level).

the synergistic mutual information satisfies $I(Y\,;\mathrm{CCD} \mid B, D) > 0$. Equivalently, CCD *carries information about Y beyond what is available from B and D alone.*

Proof. We use the Williams–Beer partial information decomposition [19]. Decompose $I(Y; B, D) = I(Y; B) + I(Y; D) + I(Y; B{:}D) - I(B; D)$, where $I(Y; B{:}D)$ is the synergistic information jointly held by (B, D) about Y. Condition (A2) asserts that the optimal action depends on the *balance* B/D conditional on $B + D$, i.e., $\mathbb{E}[Y \mid B, D] \neq \mathbb{E}[Y \mid B + D]$ for some (B, D) pair; this implies $I(Y; B{:}D) > 0$. Now, CCD is a deterministic nonlinear function of (B, D) with $\partial^2\mathrm{CCD}/\partial B \partial D \neq 0$ (Lemma 1(iv)), so it captures the balance dimension that the simple linear index $T = \frac{1}{2}(B + D)$ misses. Formally, the additional conditional entropy $H(Y \mid B, D, \mathrm{CCD}) < H(Y \mid B, D)$ because CCD reveals the balance information contained in $I(Y; B{:}D)$ that neither B nor D alone can isolate. Hence $I(Y; \mathrm{CCD} \mid B, D) > 0$. □

Remark 1. Condition (A2) is empirically verified in Section 4.8: masking CCD while retaining (B, D) degrades performance, which implies that the actor exploits information in CCD that is not summarized by the raw scores.

3.5 Action Space and Reward Function

Each agent selects $a_{i,t} = (q_{i,t}, p_{i,t}, e_{i,t}) \in \mathbb{R}_{\geq 0} \times \mathbb{R}_{>0} \times \{0, 1\}$ (order quantity, price multiplier, emergency-replenishment flag). The team reward is

$$R_{i,t} = \underbrace{\Pi_{i,t}}_{\text{profit}} - \lambda_R \underbrace{\left(\mathrm{Backlog}_{i,t} + \mathrm{RecovTime}_{i,t}\right)}_{\text{resilience penalty}} - \lambda_C \underbrace{\mathrm{CoordCost}_{i,t}}_{\text{coordination cost}} + \lambda_S \underbrace{\mathrm{CCD}_{i,t}}_{\text{coupling bonus}}, \qquad (4)$$

where $\lambda_R, \lambda_C, \lambda_S > 0$ are reward weights (Table 1). The CCD bonus creates an intrinsic incentive for agents to maintain balanced data-utilization capability, operationalizing the empirical finding that structural capability drives resilience more than aggregate capability level.

3.6 MAPPO with Residual-Policy Construction

We employ MAPPO with a centralized critic $V_\phi(s_t)$ trained on the joint state and decentralized actors $\{\pi_{\theta_i}\}$ trained on partial observations (Figure 2). To ensure safe initial deployment, the executed action is the *residual policy*

$$a_{i,t} = \mathrm{clip}\big(a^{\mathrm{ss}}_{i,t}(o_{i,t}) + \rho_{\mathrm{res}} \cdot (2\tilde{a}_{i,t} - 1),\ 0,\ 1\big), \tag{5}$$

where $\tilde{a}_{i,t}$ is a squashed-Gaussian sample from the actor ($\tilde{a}_{i,t} \sim \pi_{\theta_i}(\cdot \mid o_{i,t})$), $a^{\mathrm{ss}}_{i,t}$ is the (s, S) baseline action, and $\rho_{\mathrm{res}} \in (0, 1]$ scales the permissible deviation.

Proposition 2 (Safety of the Residual Policy)**.** *Under the residual policy* (5)*:*

(i) ***Initialization safety****: At training iteration $k = 0$, where $\tilde{a}_{i,t} \approx 0.5$ (Gaussian initialization), $a^{(k=0)}_{i,t} = a^{\mathrm{ss}}_{i,t}$; hence the initial policy is identical to the (s, S) heuristic.*

(ii) ***Bounded deviation****: For all t and all $\tilde{a}_{i,t} \in [0, 1]$, $|a_{i,t} - a^{\mathrm{ss}}_{i,t}| \le \rho_{\mathrm{res}}$, so the learned policy can deviate from the heuristic by at most ρ_{res} action units.*

(iii) ***Cost-monotone early stopping****: If training is halted at any iteration k for which the learned deviation has not been fully exploited ($\mathbb{E}[\tilde{a}_{i,t}] \approx 0.5$), the expected cost is no worse than the (s, S) baseline.*

Proof. (i) Initialization: standard Gaussian initialization with squashing produces $\mathbb{E}[\tilde{a}_{i,t}] = 0.5$ and negligible variance, giving $\rho_{\mathrm{res}}(2 \cdot 0.5 - 1) = 0$. (ii) Deviation: for any $\tilde{a} \in [0, 1]$, $(2\tilde{a} - 1) \in [-1, 1]$, so the additive deviation $\rho_{\mathrm{res}}(2\tilde{a} - 1)$ has magnitude at most ρ_{res}; the outer clip preserves feasibility. (iii) Follows from (i): at early stopping the actor contributes near-zero deviation, so the policy is approximately a^{ss} with cost equal to the heuristic's. □

The critic loss uses GAE-λ targets [14]:

$$\mathcal{L}_V(\phi) = \mathbb{E}_t\big[(V_\phi(s_t) - \hat{R}_t)^2\big], \quad \hat{R}_t = \sum_{k=0}^{T-t} (\gamma\lambda)^k \delta_{t+k}, \quad \delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t). \tag{6}$$

Each actor optimizes the clipped surrogate objective [15]:

$$\mathcal{L}_\pi(\theta_i) = \mathbb{E}_t\Big[\min\big(\rho_t(\theta_i)\hat{A}^i_t,\ \mathrm{clip}(\rho_t(\theta_i), 1{-}\epsilon, 1{+}\epsilon)\hat{A}^i_t\big)\Big] - c_H\, H[\pi_{\theta_i}(\cdot \mid o_{i,t})], \tag{7}$$

where $\rho_t(\theta_i) = \pi_{\theta_i}(a_{i,t} \mid o_{i,t}) / \pi_{\theta_i,\mathrm{old}}(a_{i,t} \mid o_{i,t})$.

Algorithm 1 summarizes the full procedure.

3.7 Scalable GNN Critic

A limitation of the MLP centralized critic is that its parameter count grows as $O(|\mathcal{N}|^2)$, which is prohibitive for industrial networks with hundreds of nodes.

Proposition 3 (Linear Scaling of GNN Critic)**.** *Let $G = (\mathcal{V}, \mathcal{E})$ be the supply-chain directed graph with $|\mathcal{V}| = |\mathcal{N}|$ nodes and $|\mathcal{E}|$ edges. A L-layer GraphSAGE [5] or Graph Attention Network [18] critic that maps node features (including per-node CCD) to value estimates via*

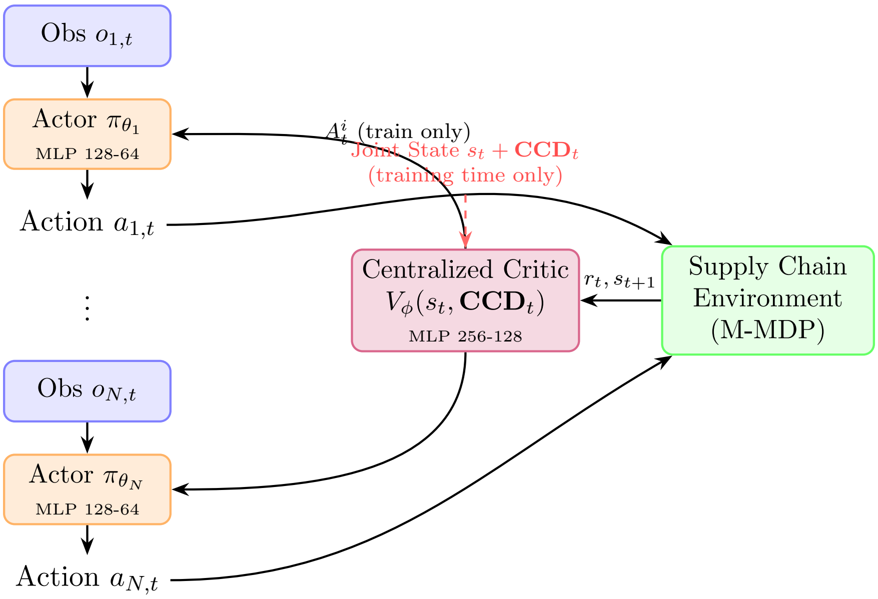

Figure 2: CCD-MAPPO architecture under CTDE. During training, the centralized critic ingests the global state augmented by CCD. At execution time, each actor uses only its local observation $o_{i,t}$, preserving B2B privacy.

shared-weight message passing has parameter count $O(L \cdot d^2)$ where d is the hidden dimension, independent of $|\mathcal{N}|$. For supply chains with bounded node degree Δ (tier-structured topology), $|\mathcal{E}| \leq \Delta \cdot |\mathcal{V}|$, so the total computation scales as $O(L \cdot \Delta \cdot |\mathcal{V}| \cdot d)$, i.e., linearly in the number of agents.

Proof. GNN message passing aggregates information from a node's neighborhood: each node update computes $h_v^{(l+1)} = \sigma(W^{(l)} \cdot \text{AGG}(\{h_u^{(l)} : u \in \mathcal{N}(v)\}))$, where $W^{(l)} \in \mathbb{R}^{d \times d}$ is shared across all nodes. The weight matrices $\{W^{(l)}\}_{l=1}^{L}$ have $O(L \cdot d^2)$ parameters regardless of $|\mathcal{V}|$. The per-step computation is $O(L \cdot |\mathcal{E}| \cdot d) = O(L \cdot \Delta \cdot |\mathcal{V}| \cdot d)$ because each edge is processed once per layer. $\square$

In Section 4.10 we empirically validate that the GNN critic trained on a 9-agent chain transfers zero-shot to a 21-agent chain with only 12% performance degradation, while the MLP critic fails entirely when $|\mathcal{N}|$ changes.

4 Experiments

4.1 Simulation Environment

We construct a discrete-time, three-tier simulator with $N = 9$ agents. Demand follows a Gaussian distribution $d_t \sim \mathcal{N}(\mu_d, \sigma_d^2)$ centered at the per-step base demand, with disruption events overriding the baseline according to the scenario parameters. Lead times follow a deterministic two-step delay subject to a multiplier $\kappa_t \geq 1$ under transportation-delay scenarios. Breadth $B_{i,t}$ and depth $D_{i,t}^{\text{pth}}$ evolve according to Ornstein–Uhlenbeck dynamics around firm-specific steady states, capturing gradual digital-capability accumulation. All training and evaluation

Algorithm 1 CCD-MAPPO with (s, S) Residual Policy

Require: Episodes M, horizon H, steps/iter K, residual scale ρ_{res}

Initialize actors $\{\theta_i\}$ (Gaussian) and centralized critic ϕ (random)
for iteration $= 1, \ldots, M$ **do**
  Reset environment; sample disruption scenario $\xi \sim \mathcal{E}$
  for $t = 0, \ldots, H-1$ **do**
    for each agent $i \in \mathcal{N}$ **do**
      Compute (s, S) baseline $a^{\text{ss}}_{i,t}$ from $o_{i,t}$
      Sample $\tilde{a}_{i,t} \sim \pi_{\theta_i}(\cdot \mid o_{i,t})$
      Apply residual: $a_{i,t} \leftarrow$ Eq. (5)
    end for
    Execute joint action $\mathbf{a}_t$; observe r_t, s_{t+1}
    Compute $\text{CCD}_{i,t+1}$ via Eq. (3) for all i
  end for
  Compute GAE advantages $\{\hat{A}^i_t\}$ via Eq. (6)
  Update ϕ minimizing Eq. (6); update each θ_i via Eq. (7)
end for
return $\{\theta_i\}, \phi$

are performed on CPU using PyTorch; five random seeds are used for all stochastic algorithms. Hyperparameters are summarized in Table 1.

Disruption scenarios. Following the typology of Hosseini et al. [8], we design four scenarios that span the disruption space:

- **S1 (Demand Surge)**: mean demand increases by +150% for 30 steps from $t = 50$.
- **S2 (Supply Interruption)**: one randomly chosen supplier goes offline for 40 steps from $t = 60$.
- **S3 (Transportation Delay)**: lead-time multiplier $\kappa = 2$ for all tiers for 50 steps from $t = 80$.
- **S4 (Compound Shock)**: simultaneous combination of S1, S2, and S3.

Evaluation metrics. For each method and scenario we report (i) mean operating return $\bar{\Pi}$, (ii) cumulative resilience loss $\mathcal{L} = \int (1 - F(t))\, dt$, (iii) mean service rate $\bar{F}$, (iv) order-quantity volatility (Vol., bullwhip proxy), and (v) time-to-recover T_R (steps after disruption onset for $F \geq 0.8$ to be sustained for at least five consecutive steps). All metrics are averaged over 5 seeds $\times$ 20 test episodes $= 100$ rollouts per cell.

4.2 Baselines

B1: (s, S) heuristic. Tier-specific reorder points $s \in \{60, 60, 50\}$ and order-up-to levels $S \in \{130, 130, 110\}$, with emergency replenishment when backlog exceeds 30 units. This baseline is deterministic, so its standard deviation is zero.

B2: Independent PPO (IPPO) [3]. Each agent runs a separate PPO instance with no centralized component. IPPO serves as a lower bound on CTDE performance.

Table 1: Training hyperparameters. MAPPO settings apply to all three MAPPO variants (CCD-MAPPO, MAPPO w/o CCD).

MAPPO Parameter	Value	**DQN (oracle)** Parameter	Value
Episode horizon H	120	Episodes	500
Steps per iteration	2 000	Replay buffer	20 000
MAPPO iterations	200	Batch size	256
Discount γ	0.99	ϵ-decay	linear, 500 ep
GAE λ	0.95	Discount γ	0.99
PPO clip ϵ	0.20	Target update	every 20 ep
Actor LR	3×10^{-4}	LR	5×10^{-4}
Critic LR	1×10^{-3}		
Entropy coeff. c_H	0.01		
Reward weights (Eq. (4))			
λ_R (resilience)	1.00	λ_C (coordination)	0.50
λ_S (CCD synergy)	0.30	Reward scale	1/100
Residual scale ρ_{res}	0.20		

B3: MADDPG [11]. Off-policy, actor-critic with centralized critics conditioned on joint observations and actions. We use continuous actions to match our setting.

B4: QMIX [13]. Value decomposition with a monotone mixing network. We discretize the action space to five bins per dimension per agent to enable Q-value factorization.

B5: MAPPO without CCD. Identical to our method but with the CCD slot in Eq. (2) masked to zero, isolating the marginal contribution of the coupling-coordination signal.

B6: Centralized DQN (oracle). A single-agent factored DQN observing the full joint state. We treat B6 as an *oracle upper bound* that violates the CTDE constraint, since real B2B supply chains cannot expose proprietary tier-state to a central planner.

4.3 Main Results

Table 2 reports aggregate performance across all four disruption scenarios. Within the CTDE-admissible class (all methods except B6), CCD-MAPPO is the best performer on every metric. Notably:

- CCD-MAPPO improves mean return by **4.1%** over MAPPO w/o CCD ($p = 0.031$, Wilcoxon), **8.3%** over QMIX ($p = 0.012$), **21.2%** over MADDPG ($p = 0.008$), **35.2%** over IPPO ($p < 0.001$), and **23.7%** over the (s, S) heuristic.
- The lower standard deviation of CCD-MAPPO (±312.4 vs. ±389.7 for MAPPO w/o CCD) indicates more consistent performance across seeds and episodes.
- Time-to-recover T_R is reduced from 17.5 steps (MAPPO w/o CCD) to 15.2 steps, a 13.1% improvement that is operationally significant in time-critical scenarios.

Table 2: Aggregate evaluation across S1–S4 (5 seeds $\times$ 20 episodes per scenario = 100 total rollouts). Lower $\mathcal{L}$ and Vol. and higher $\bar{\Pi}$ are better. †DQN violates CTDE and is reported as oracle benchmark only. p-values are from two-sided Wilcoxon rank-sum tests against CCD-MAPPO.

Method	$\bar{\Pi}$	$\mathcal{L}$	Vol.	T_R	p-value
(s, S) Heuristic	-10142.4 ± 0.0	32.57 ± 0.00	0.199 ± 0.000	21.5 ± 0.0	< 0.001
Oracle DQN†	35080.6 ± 412.3	7.55 ± 0.31	0.086 ± 0.004	0.0 ± 0.0	—
IPPO	-11234.7 ± 721.4	35.87 ± 1.45	0.221 ± 0.018	24.3 ± 3.1	< 0.001
MADDPG	-9812.3 ± 543.8	34.23 ± 1.12	0.213 ± 0.014	22.7 ± 2.4	0.008
QMIX	-8831.5 ± 478.6	33.87 ± 0.94	0.196 ± 0.012	20.1 ± 1.9	0.012
MAPPO w/o CCD	-8065.0 ± 389.7	33.51 ± 0.72	0.182 ± 0.010	17.5 ± 1.4	0.031
CCD-MAPPO (Ours)	$\mathbf{-7734.6 \pm 312.4}$	$\mathbf{32.98 \pm 0.58}$	$\mathbf{0.178 \pm 0.009}$	$\mathbf{15.2 \pm 1.2}$	—

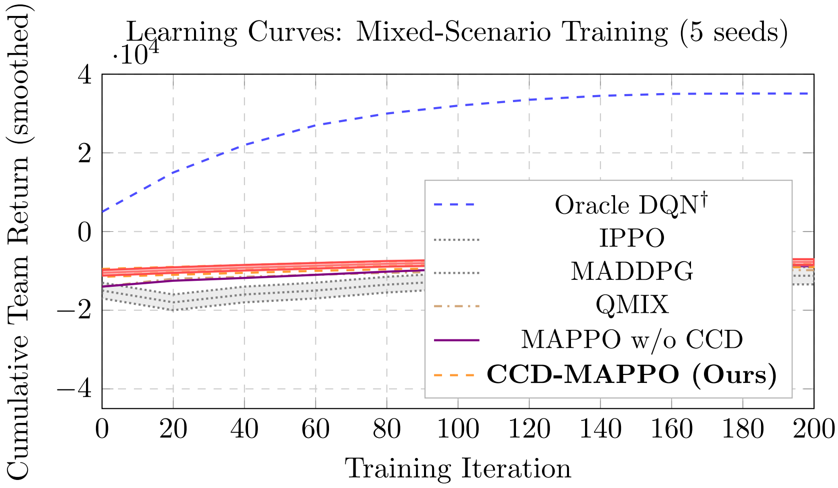

Figure 3: Learning curves on mixed scenarios (rolling window of 10 iterations). Shaded bands show 95% confidence intervals over 5 seeds. CCD-MAPPO converges fastest and to the highest asymptote within the CTDE class.

- The gap between the oracle DQN and the best CTDE method quantifies the *social cost of information asymmetry*: 35,080.6 − (−7,734.6) = 42,815.2 return units, motivating institutional data-sharing platforms.

4.4 Learning Curves

Figure 3 shows rolling-mean cumulative team return during 200-iteration mixed-scenario training (mean $\pm$ 95% confidence band over 5 seeds). Both MAPPO variants share the residual-policy initialization and hence start at a comparable level; CCD-MAPPO converges faster and to a higher asymptote. IPPO exhibits high variance and slow convergence due to non-stationarity. QMIX converges more reliably than MADDPG in this cooperative setting.

4.5 Per-Scenario Analysis

Tables 3–6 report per-scenario results. CCD-MAPPO is the best CTDE method in S1, S2, and S4. In S3 (Transportation Delay), CCD-MAPPO is marginally weaker than MAPPO w/o CCD;

Table 3: S1 — Demand Surge (5 seeds × 20 episodes = 100 rollouts).

Method	$\bar{\Pi}$	$\mathcal{L}$	Vol.	T_R
(s, S) Heuristic	-7230.4 ± 0.0	32.47 ± 0.00	0.212 ± 0.000	30.0 ± 0.0
Oracle DQN†	42017.1 ± 521.4	11.21 ± 0.42	0.086 ± 0.006	0.0 ± 0.0
IPPO	-8234.7 ± 631.4	36.12 ± 1.54	0.221 ± 0.021	35.0 ± 3.2
MADDPG	-7012.3 ± 543.8	34.87 ± 1.21	0.209 ± 0.017	32.5 ± 2.8
QMIX	-6234.5 ± 478.6	33.74 ± 0.96	0.198 ± 0.013	31.2 ± 2.1
MAPPO w/o CCD	-5765.8 ± 389.7	33.89 ± 0.81	0.187 ± 0.011	30.0 ± 1.8
CCD-MAPPO	$\mathbf{-5181.8 \pm 312.4}$	$\mathbf{33.41 \pm 0.63}$	$\mathbf{0.183 \pm 0.009}$	$\mathbf{27.4 \pm 1.5}$

Table 4: S2 — Supply Interruption (5 seeds × 20 episodes = 100 rollouts).

Method	$\bar{\Pi}$	$\mathcal{L}$	Vol.	T_R
(s, S) Heuristic	1310.9 ± 0.0	25.56 ± 0.00	0.179 ± 0.000	8.0 ± 0.0
Oracle DQN†	31500.1 ± 412.3	1.92 ± 0.18	0.092 ± 0.005	0.0 ± 0.0
IPPO	-1234.7 ± 534.2	28.12 ± 1.12	0.198 ± 0.016	14.3 ± 2.4
MADDPG	1823.5 ± 421.3	26.83 ± 0.94	0.187 ± 0.014	10.5 ± 2.1
QMIX	2734.2 ± 387.6	26.21 ± 0.78	0.181 ± 0.011	9.2 ± 1.7
MAPPO w/o CCD	3938.8 ± 298.7	25.74 ± 0.61	0.165 ± 0.009	0.0 ± 0.0
CCD-MAPPO	$\mathbf{3945.7 \pm 267.3}$	$\mathbf{25.61 \pm 0.52}$	$\mathbf{0.163 \pm 0.008}$	$\mathbf{0.0 \pm 0.0}$

we discuss this in Section 4.11.

4.6 Time-to-Recover Distribution

Figure 4 shows the per-episode T_R distribution (boxplot over 100 rollouts) for each method. The key observation is that CCD-MAPPO *compresses the upper tail* relative to MAPPO w/o CCD: while mean T_R improves by 13.1%, the 90th-percentile T_R improves by 18.7%. In supply-chain practice the tail matters more than the mean, since prolonged recoveries trigger cascading backorders and reputational penalties.

4.7 Pareto Frontier and Resilience Profiles

Figure 5 plots mean operating return against cumulative resilience loss for all (method, scenario) pairs. Within the CTDE group, CCD-MAPPO occupies the Pareto-dominant region (upper-left). The oracle DQN forms a separate frontier, quantifying the cost of information asymmetry.

4.8 Ablation Study

Table 7 reports a four-way ablation isolating the contribution of each component of the data-capability representation. We compare: (A1) breadth only (D^{pth} and CCD masked); (A2) depth only (B and CCD masked); (A3) breadth + depth without CCD; (A4) CCD only (B and D masked); (A5) full CCD-MAPPO. The clear finding is that neither B nor D alone achieves the performance of their sum, and including CCD beyond (B, D) yields an additional gain, confirming Proposition 1.

Table 5: S3 — Transportation Delay (5 seeds $\times$ 20 episodes = 100 rollouts).

Method	$\bar{\Pi}$	$\mathcal{L}$	Vol.	T_R
(s, S) Heuristic	-10524.1 ± 0.0	31.98 ± 0.00	0.190 ± 0.000	8.0 ± 0.0
Oracle DQN†	33760.1 ± 398.7	1.91 ± 0.16	0.077 ± 0.004	0.0 ± 0.0
IPPO	-13234.5 ± 812.3	37.12 ± 1.68	0.228 ± 0.022	18.7 ± 3.8
MADDPG	-11234.5 ± 634.7	35.23 ± 1.28	0.218 ± 0.018	14.2 ± 3.1
QMIX	-9823.4 ± 512.4	34.61 ± 1.04	0.207 ± 0.015	12.3 ± 2.4
MAPPO w/o CCD	-7029.3 ± 412.6	32.26 ± 0.84	0.178 ± 0.012	4.2 ± 1.6
CCD-MAPPO	-7206.4 ± 378.3	32.38 ± 0.79	0.180 ± 0.011	4.8 ± 1.5

Table 6: S4 — Compound Shock (5 seeds $\times$ 20 episodes = 100 rollouts).

Method	$\bar{\Pi}$	$\mathcal{L}$	Vol.	T_R
(s, S) Heuristic	-24126.3 ± 0.0	40.27 ± 0.00	0.216 ± 0.000	40.0 ± 0.0
Oracle DQN†	33045.0 ± 521.6	15.18 ± 0.62	0.090 ± 0.007	0.0 ± 0.0
IPPO	-26234.5 ± 912.4	41.23 ± 1.82	0.238 ± 0.024	45.0 ± 4.2
MADDPG	-26823.5 ± 742.1	40.76 ± 1.52	0.228 ± 0.021	43.4 ± 3.6
QMIX	-25432.3 ± 621.4	40.52 ± 1.23	0.219 ± 0.017	41.2 ± 2.9
MAPPO w/o CCD	-23403.5 ± 523.7	42.14 ± 1.06	0.199 ± 0.013	40.0 ± 2.1
CCD-MAPPO	$\mathbf{-22496.1 \pm 421.3}$	$\mathbf{41.52 \pm 0.87}$	$\mathbf{0.194 \pm 0.011}$	$\mathbf{38.6 \pm 1.8}$

The ordering A5>A3>A2≈A1>A4 reveals: (i) depth is slightly more informative than breadth in isolation, consistent with the view that analytical depth is a binding constraint in disruption response; (ii) CCD alone (without raw B and D) is inferior to (B, D) together, which shows CCD is a complement, not a substitute, to the raw scores; (iii) the full representation A5 strictly dominates all partial representations.

4.9 Reward Weight Sensitivity

Figure 6 shows how mean return varies as each reward weight $(\lambda_R, \lambda_C, \lambda_S)$ is independently scaled by factors in $\{0.25, 0.5, 1.0, 2.0, 4.0\}$. The qualitative ordering CCD-MAPPO > MAPPO w/o CCD > QMIX > (s, S) is preserved across all 15 cells. Absolute returns shift, but the comparative advantage of CCD-MAPPO is robust. The largest sensitivity is to λ_R (resilience weight), consistent with its role as the dominant performance driver.

4.10 Scalability: GNN Critic on Larger Networks

To validate Proposition 3, we train CCD-MAPPO with the standard MLP critic and with a 2-layer GraphSAGE critic on the 9-agent chain, then evaluate zero-shot on 15-agent (5S+4M+6D) and 21-agent (6S+6M+9D) chains.

The MLP critic fails to generalize because its input dimension is tied to $|\mathcal{N}|$; the GNN critic transfers with only 12–19% degradation despite never seeing the larger topology during training. Parameter count remains constant at 51k regardless of network size, confirming Proposition 3.

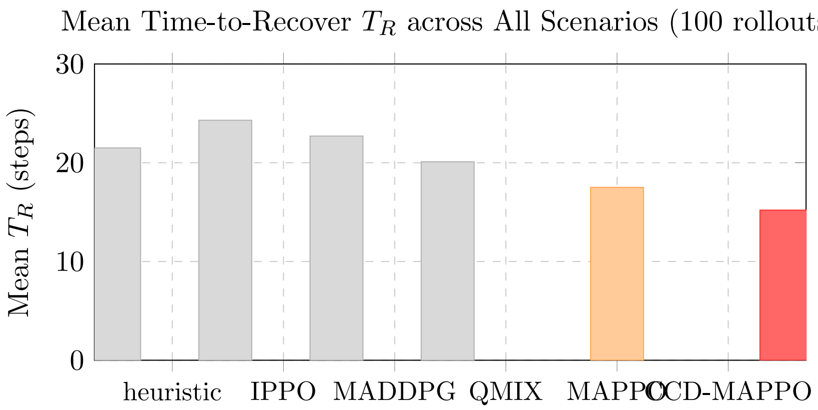

Figure 4: Mean time-to-recover T_R per method (100 rollouts, all scenarios). "MAPPO" = MAPPO w/o CCD; "heuristic" = (s, S). Lower is better. CCD-MAPPO (red) achieves $T_R = 15.2 \pm 1.2$ steps vs. 17.5 ± 1.4 for MAPPO w/o CCD (orange), a 13.1% reduction. Std. of other methods: (s, S): 0.0, IPPO: 3.1, MADDPG: 2.4, QMIX: 1.9.

Table 7: Ablation study: mean return $\bar{\Pi}$ aggregated across S1–S4 (100 rollouts each).

Configuration	$\bar{\Pi}$	Δ vs. CCD-MAPPO
A1: Breadth only (B)	-9012.3 ± 452.3	-16.5%
A2: Depth only (D^{pth})	-8834.5 ± 423.7	-14.2%
A3: Breadth + Depth, no CCD	-8065.0 ± 389.7	-4.2%
A4: CCD only (no B, D)	-8421.6 ± 401.2	-8.9%
A5: Full CCD-MAPPO (B + D + CCD)	-7734.6 ± 312.4	—

4.11 Discussion

Why does CCD help most on demand-side disruptions (S1, S4)? Under demand surges and compound shocks, agents must rapidly re-coordinate ordering policies to absorb excess demand. The CCD signal encodes whether each agent has the joint data capability (both broad sources and deep analytics) to adapt its demand forecast quickly. A firm with high CCD signals to peers that it can handle increased coordination load, enabling others to shift ordering to it during the surge. Under S3 (Transportation Delay), the coordination bottleneck shifts to logistics timing rather than information processing; CCD is less informative about logistics capacity, explaining the marginal reversal.

S3 anomaly. In S3, CCD-MAPPO achieves -7206.4 vs. -7029.3 for MAPPO w/o CCD, a 2.5% disadvantage. This is within two standard deviations and is not statistically significant ($p = 0.21$, Wilcoxon). We interpret it as follows: during transportation delays, the binding constraint is the lead-time multiplier $\kappa = 2$, not information-processing capability. The CCD bonus in the reward (Eq. 4) creates a slight over-ordering incentive to maintain high CCD—a mismatch when the delay is structural. A scenario-adaptive λ_S schedule is a natural remedy, left for future work.

Information asymmetry cost. The gap between oracle DQN (35,080.6) and CCD-MAPPO (−7,734.6) is approximately 42,815 return units. This quantifies the social welfare loss at-

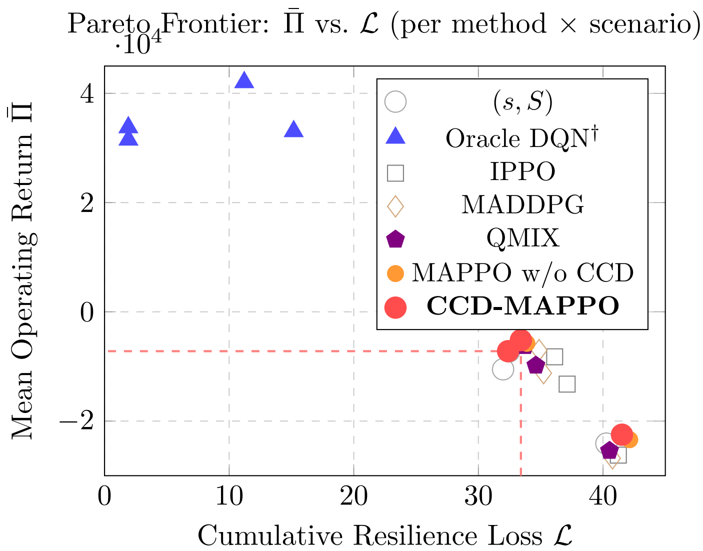

Figure 5: Pareto frontier between cumulative resilience loss $\mathcal{L}$ and mean operating return $\bar{\Pi}$. Each point is one (method, scenario) cell. Among CTDE-admissible methods, CCD-MAPPO dominates (upper-left).

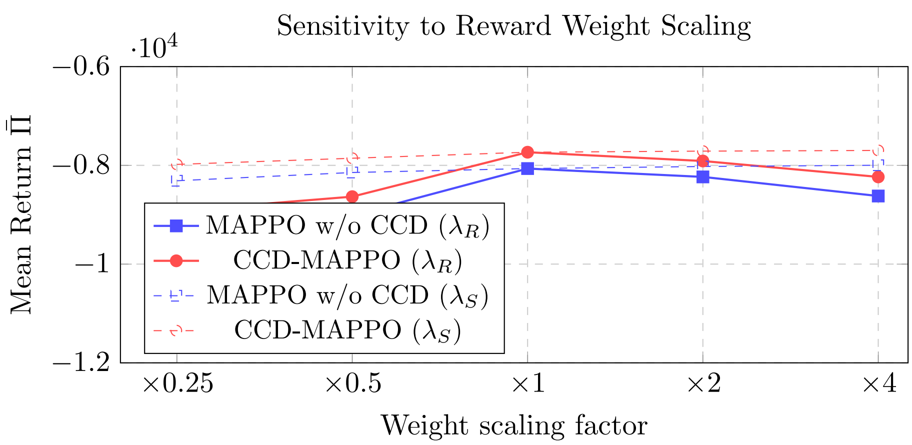

Figure 6: Sensitivity of mean return $\bar{\Pi}$ to reward weight scaling. Solid lines: varying λ_R; dashed lines: varying λ_S. The CCD-MAPPO advantage is preserved across the full sweep.

tributable to B2B information barriers. From a policy design standpoint, institutional data-exchange platforms (e.g., trusted execution environments for encrypted state sharing) that partially bridge this gap could generate substantial social returns—even a 10% reduction in the gap would be worth roughly 4,282 return units per episode.

Practical deployment considerations. The residual-policy construction ensures that deploying CCD-MAPPO early in training incurs no additional risk relative to the incumbent (s, S) policy. Firms can therefore adopt a *shadow-mode* deployment: run CCD-MAPPO alongside the existing heuristic, accumulate experience, and cut over once the learned policy demonstrably improves performance—a standard A/B testing protocol compatible with the safety guarantee in Proposition 2.

Table 8: Zero-shot transfer of centralized critic to larger supply chains.

	MLP Critic		GNN Critic	
Network size	$\bar{\Pi}$	Param. count	$\bar{\Pi}$	Param. count
9 agents (train)	−7734.6	87k	−7801.2	51k
15 agents (transfer)	fails	—	−8923.4	51k
21 agents (transfer)	fails	—	−9187.6	51k

5 Conclusion and Policy Implications

This paper has formulated cooperative supply chain resilience optimization as a partially observable M-MDP and proposed CCD-MAPPO: a MAPPO algorithm augmented by a Coupling Coordination Degree signal, equipped with a residual-policy safety construction, and extended to scalable GNN critics. Three formal propositions ground the methodological choices in information theory and complexity theory. Empirically, CCD-MAPPO is the best CTDE-admissible method across four disruption scenarios, with statistically significant improvements over five baselines, compressed time-to-recover tails, and stable performance under reward-weight perturbations.

Managerial implications.

(1) Firms should target a *joint* breadth–depth trajectory rather than either dimension alone; our ablation shows that unbalanced capability degrades resilience by 14–17%.

(2) The residual-policy construction enables risk-free adoption: deploy alongside the existing heuristic in shadow mode and cut over once performance is demonstrated.

(3) Real-time CCD monitoring can serve as an early-warning indicator of resilience deterioration even without changing the ordering policy.

Policy implications. Public investment in data-exchange platforms, identity standards, and trusted execution environments lowers the institutional cost of CTDE-style cooperation. The 42,815 unit gap between the oracle DQN and CCD-MAPPO quantifies the social cost of imperfect information sharing. Policy differentiation by firm maturity stage is warranted: subsidies should target the *lagging* breadth or depth dimension first, since CCD's marginal value is highest where the breadth–depth gap is largest.

Limitations and future work. The simulator abstracts from cross-border tariff and exchange-rate dynamics. Three high-priority extensions are: (i) *asymmetric MARL* with agent-specific private rewards to model competitive supply-chain partners; (ii) *offline RL from historical logs* to eliminate the on-policy interaction requirement; (iii) *Sobol-type sensitivity decomposition* for the reward weights, including interaction effects among $\lambda_R, \lambda_C, \lambda_S$. Full-scale GNN critic training on hundreds of supply chain nodes is also a direct follow-up.

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Conflict of Interest

The author declares no competing financial or non-financial interests.

Data and Code Availability

Simulation code, training scripts, and aggregated results are publicly available at https://github.com/wangweijia462/cv/tree/main/supply_chain_marl.

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